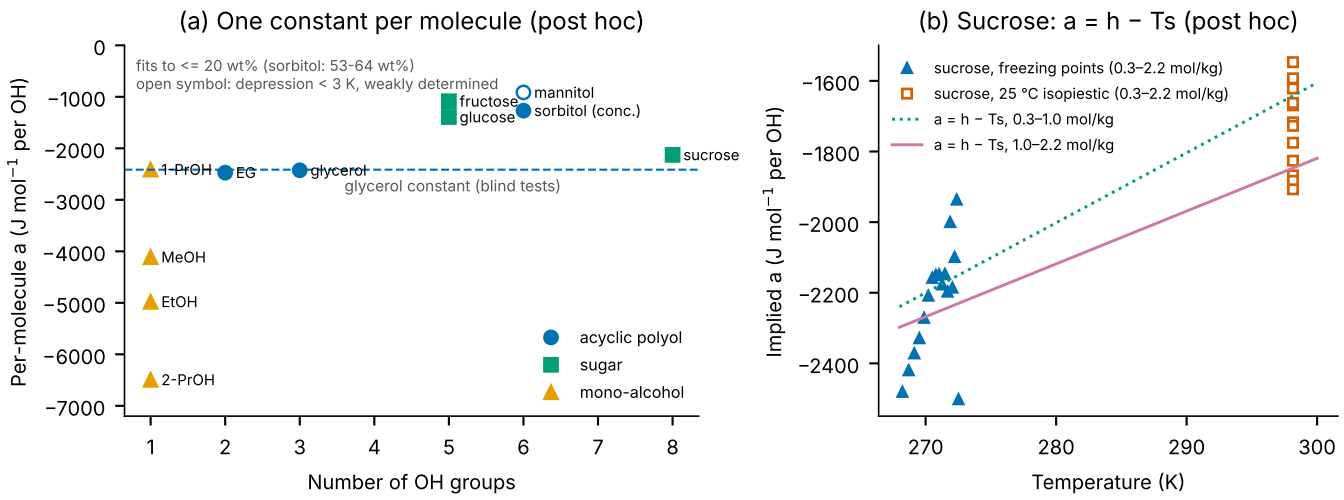
(a) One constant per molecule (post hoc)
fits to <= 20 wt% (sorbitol: 53-64 wt%)
open symbol: depression < 3 K, weakly determined
mannitol
fructose
glucose
sorbitol (conc.)
sucrose
1-PrOH
EG
glycerol
glycerol constant (blind tests)
MeOH
EtOH
2-PrOH
acyclic polyol
sugar
mono-alcohol
Per-molecule a (J mol⁻¹ per OH)
Number of OH groups
0
−1000
−2000
−3000
−4000
−5000
−6000
−7000
1
2
3
4
5
6
7
8
(b) Sucrose: a = h − Ts (post hoc)
sucrose, freezing points (0.3–2.2 mol/kg)
sucrose, 25 °C isopiestic (0.3–2.2 mol/kg)
a = h − Ts, 0.3–1.0 mol/kg
a = h − Ts, 1.0–2.2 mol/kg
Implied a (J mol⁻¹ per OH)
Temperature (K)
−1600
−1800
−2000
−2200
−2400
270
280
290
300